



COMPUTER SECURITY

2025

FINAL BANK (Unsolved)



SENIOR
25

BY AB & AE & AM & MR

Join our Telegram!
@qahwa_is



Choose the correct answer:

1.	The need for keeping information secret.						
	a)	Confidentiality	b)	Integrity	c)	Availability	d)
2.	Trustworthiness of data and resources.						
	a)	Confidentiality	b)	Integrity	c)	Availability	d)
3.	The ability to use the information or resource						
	a)	Confidentiality	b)	Integrity	c)	Availability	d)
4.	Timely response is an expectation of _____ security goal						
	a)	Integrity	b)	Availability	c)	Confidentiality	d)
5.	_____ means that some unauthorized party (person, program) has gained access to data						
	a)	Interception	b)	Modification	c)	Interruption	d)
6.	_____ an asset of the system becomes lost, unavailable or unusable						
	a)	Interruption	b)	Interception	c)	Modification	d)
7.	_____ means that some unauthorized party (person, program) has gained access to an asset						
	a)	Interruption	b)	Interception	c)	Modification	d)
8.	_____ an additional computation, or modifying data being transmitted by the network.						
	a)	Interruption	b)	Interception	c)	Modification	d)
9.	_____ counterfeit objects on a computing system. (adding records to an existing data base or insertion of spurious transactions to a network communication system.						
	a)	Interruption	b)	Interception	c)	Modification	d)
10.	Cryptography is used as a defense technique against the following attacks except _____						
	a)	Modification	b)	Interception	c)	Interruption	d)
11.	Erasure of program or data or malfunction of an OS file manager is considered _____						
	a)	Modification	b)	Interception	c)	Interruption	d)
12.	Flooding is an attack of _____						
	a)	Integrity	b)	Availability	c)	Confidentiality	d)
13.	Spoofing is an attack of _____						
	a)	Integrity	b)	Availability	c)	Confidentiality	d)
14.	_____ is when one host pretends to be another						
	a)	Eavesdrop	b)	Spoofing	c)	Wiretap	d)
15.	When a host pretends to be another, for example, using his IP's, is known as _____ attack						
	a)	Eavesdrop	b)	Masquerade	c)	Wiretap	d)
16.	_____ is intercepting and carrying on a session begun by another entity						
	a)	Masquerade	b)	Impersonation	c)	Session Hijacking	d)



17.	_____ in which one entity intrudes between two others.							
	a)	Masquerade	b)	Impersonation	c)	Session Hijacking	d)	Man-in-the middle
18.	A sender should not be able falsely to deny later that he/she sent a message							
	a)	Integrity	b)	Confidently	c)	Privacy	d)	Non-repudiation
19.	Which security mechanism ensures that a user cannot deny their actions?							
	a)	Authentication	b)	Authorization	c)	Confidentiality	d)	Non-repudiation
20.	Program controls include:							
	a)	Peer review	b)	Proof of program correctness	c)	Independent testing	d)	All
21.	Legal and ethical controls are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Policy controls	d)	Physical controls
22.	Cyber security rules for daily practicing are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Administration controls	d)	Physical controls
23.	Eavesdropping and impersonate are attacks of _____ Respectively							
	a)	Availability , Confidentiality	b)	Availability , Integrity	c)	Confidentiality , Integrity	d)	Integrity , Confidentiality
24.	Choose the correct sentence?							
	a)	Erasure of a program is considered an interruption threat						
	b)	TOCTOU is considered an attack of availability						
	c)	Flooding are attacks of confidentiality						
	d)	Error is a mistake in interpreting a syntax error in a piece of code						
25.	_____ Consuming processor capacity, system memory, disk space							
	a)	Bacteria	b)	Resident	c)	Virus	d)	Worm
26.	_____ Attaches itself to program and propagates copies of itself to other programs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
27.	_____ Contains unexpected, additional functionality							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
28.	_____ Triggers action when condition occurs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
29.	_____ Triggers action when specified time occurs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
30.	_____ Allows unauthorized access to functionality							
	a)	Virus	b)	Trapdoor	c)	Worm	d)	Rabbit
31.	_____ Propagates copies of itself through a network							
	a)	Virus	b)	Trapdoor	c)	Worm	d)	Rabbit



32.	_____Replicates itself without limit to exhaust resources			
	a) Virus	b) Trapdoor	c) Worm	d) Rabbit
33.	_____is a class of malicious code that "detonates" or goes off when a specified condition occurs			
	a) Virus	b) Trojan horse	c) Logic bomb	d) Time bomb
34.	The _____ is a code embedded in program that is set to explode when certain condition is/are met			
	a) Trojan house	b) Trapdoor	c) Worm	d) Logical bomb
35.	_____ replicates a copy of itself to other systems to execute a malicious code			
	a) Virus	b) Worm	c) Trojan horse	d) Trapdoor
36.	_____ is a program that spreads copies of itself through a network			
	a) Virus	b) Trapdoor	c) Worm	d) Rabbit
37.	_____is a feature in a program by which someone can access the program other than by the obvious, direct call, perhaps with special privileges			
	a) Virus	b) Trapdoor	c) Worm	d) Rabbit
38.	_____ virus runs when its attached program executes and terminates when its attached program ends			
	a) Transient	b) Resident	c) Virus	d) Worm
39.	_____ virus locates itself in memory so that it can remain active even after its attached program ends			
	a) Transient	b) Resident	c) Virus	d) Worm
40.	A type of virus that executes only once, spreading its infection and causing its effect in that one execution.			
	a) Memory resident virus	b) Boot sector virus	c) One time execution	d) Appended virus
41.	A kind of malicious software runs when a computer is started, it changes control on the OS instructions			
	a) Memory resident virus	b) Boot sector virus	c) One time execution	d) Appended virus
42.	A type of virus which is active as long as the computer running			
	a) Memory resident virus	b) Brain virus	c) Polymorphic virus	d) Logic bomb
43.	Virus is active as long as the computer running			
	a) Memory resident	b) Brain virus	c) Polymorphic virus	d) Trojan horse
44.	Viruses resident in application programs and libraries are categorized as:			
	a) One time execution virus	b) Boot sector virus	c) Other homes of virus	d) Memory resident
45.	A program virus attaches itself to a program; then, whenever the program is run, the virus is activated. This kind of attachment is usually known as _____			
	a) integrated virus	b) Surrounded virus	c) Appended virus	d) replacement virus
46.	controls against program threats include the following actions except _____			
	a) peer review process	b) Independent testing	c) Proof of correctness	d) Decapsulation of different process



47.	Each virus has a signature that could be used to detect its existence, such as:							
	a)	Its location	b)	its execution performance	c)	its replication technique	d)	All
48.	What is a weakness of Caesar cipher?							
	a)	Vulnerable to brute-force attacks	b)	Vulnerable to frequency analysis	c)	limited key space	d)	all
49.	What is the main weakness of simple substitution ciphers?							
	a)	They are slow to encrypt messages	b)	They require large keys	c)	They can be easily broken using frequency analysis	d)	They are not suitable for digital encryption
50.	What is true about Steganography?							
	a)	it is used to encrypt images and send it in unreadable forms						
	b)	The encrypted file has smaller size compared with the original one						
	c)	it can be broken by brute-force attack						
	d)	it uses images to hide important data in LSB of each image's pixel						
51.	Steganography uses _____ in graphic image to hide important data							
	a)	LSB	b)	HSB	c)	MSB	d)	NSB
52.	_____ is a mistake in interpreting a requirement or a syntax error in a piece of code							
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault
53.	_____ is a human mistake in performing some software activity that may lead to a fault in computer program							
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault
54.	Validation error , domain error , and boundary condition violation are types of _____							
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault
55.	A _____ may cause a failure (which is a departure from the system's required behavior).							
	a)	Bug	b)	Error	c)	fault	d)	failure
56.	A _____ is an inside view of the system, seen by the developers, whereas a _____ is an outside view seen by the user							
	a)	Failure, fault	b)	Fault, Failure	c)	Error, Bug	d)	Fault, Flaws
57.	What is true about program flaws?							
	a)	It could be validation error or domain error						
	b)	It could lead to failure						
	c)	It could be malicious or non malicious						
	d)	All are true						
58.	Consist of replacing the valid source and/or destination IP address with false ones							
	a)	Smurf attack	b)	Spoofing attack	c)	DoS	d)	Sniffing attack



59.	For exchanging symmetric key among 10 user, we need _____ exchange operation							
	a)	45	b)	100	c)	90	d)	1000
60.	Authentication based on Something the individual has as:							
	a)	ID	b)	Password	c)	ATM card	d)	PIN
61.	Authentication based on Something the individual knows as:							
	a)	PIN	b)	Password	c)	ATM card	d)	A & B
62.	Authentication based on Something the individual is as:							
	a)	Voice patterns	b)	Handwriting	c)	Fingerprint	d)	All
63.	Password and PIN are examples for user authentication of something _____							
	a)	The user has	b)	The user is	c)	The user knows	d)	The user
64.	What is the Row Transposition Cipher?							
	a)	A substitution cipher that rearranges the order of letters in a message						
	b)	A cipher that replaces each letter in a message with a numerical value						
	c)	A cipher that applies a mathematical formula to each character in a message						
	d)	A cipher that rearranges the order of words in a message						
65.	Which of the following is a common technique used with transposition cipher?							
	a)	Rail fence	b)	Caesar	c)	Vagenre	d)	Playfair
66.	The mechanisms used to confirm a user's identity is known as:							
	a)	Identification	b)	Authentication	c)	Authorization	d)	Access limitation
67.	Each virus has a signature that could be used to detect its existence, such as:							
	a)	Location	b)	Execution performance	c)	Spreading	d)	All are true
68.	In _____, Users interact with directory to obtain any desired public key securely							
	a)	Certificate authority	b)	Public Key authority	c)	Public announcement	d)	Public Discovery
69.	_____ allow key exchange without real-time access to public-key authority							
	a)	Certificate authority	b)	Public key authority	c)	Public announcement	d)	Public directory
70.	Example for classical substitution encryption							
	a)	Playfair	b)	DES	c)	3DES	d)	AES
71.	Triple-DES with two keys is defined by $C = EK_3(DK_2(EK_1(P)))$, if K_1 equal K_3 , the triple-DES can be considered							
	a)	Double DES	b)	Triple-DES	c)	Single-DES	d)	Useless encryption



72.	Triple-DES with two keys is defined by $C = Ek_1(Dk_2(Ek_1(P)))$, if K_1 equals K_2 , the triple-DES will be							
	a)	Double DES	b)	Triple-DES	c)	Single-DES	d)	Useless encryption
73.	Which of the following is a good description for Triple-DES?							
	a)	$C = Ek_1(Dk_2(Ek_1(P)))$	b)	$C = Ek_2(Dk_1(Ek_1(P)))$	c)	$C = Ek_3(Ek_2(Ek_1(P)))$	d)	$C = Dk_3(Dk_2(Dk_1(P)))$
74.	DES encrypt data in blocks of size _____, using a key of length _____							
	a)	64-bit , 32-bit	b)	32-bit, 32-bit	c)	32-bit , 56-bit	d)	64-bit , 56-bit
75.	What is the number of rounds in single DES?							
	a)	64	b)	56	c)	48	d)	16
76.	What is the block size of triple DES?							
	a)	64	b)	128	c)	256	d)	512
77.	_____ is an attempt to discover PW by attempting to use every possible PW from a predefined list.							
	a)	Brute-force attack	b)	Dictionary attack	c)	Sniffing attack	d)	Cracking
78.	_____ provides a centralized authentication server to authenticate users to servers and servers to users.							
	a)	Trusted Third Party (TTP)	b)	Trusted Peer Party (TPP)	c)	Trusted Party Model (TPM)	d)	None
79.	_____ is a protocol that produces the same effect as a real signature							
	a)	Digital signatures	b)	Symmetric Key	c)	Asymmetric Key	d)	None
80.	Digital signatures provide the ability to:							
	a)	verify author, date & time of signature	b)	be verified by third parties to resolve disputes	c)	authenticate message contents	d)	All are true
81.	_____ refers to a key storage facility which provides secure storage of keys for future use, e.g.							
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)	Key Recovery
82.	_____ refers to procedures by which keys for notarization or nonrepudiation services can be securely archived.							
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)	Key Recovery
83.	_____ enables parties to update securely their keying material.							
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)	Key Recovery
84.	_____ refers to cryptographic keys which may become lost due to human error, software bugs, or hardware malfunction							
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)	Key Recovery



85.	_____ refers to procedures by which parties are assured of the secure destruction of keys that are no longer needed						
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)
86.	_____ refers to the procedure by which keys or pairs of keys of good cryptographic quality are securely and unpredictably generated						
	a)	Storage of Keying Material	b)	Key Archival	c)	Key Replacement	d)
87.	_____ provide some link between an entity and its uniquely defined keys						
	a)	Entity Registration	b)	Key Replacement	c)	Entity Authentication	d)
88.	_____ is to corroborate that an entity is the one claimed						
	a)	Key Replacement	c)	Entity Authentication	d)	Entity Replacement	d)
89.	_____ is the registration of messages to attest at a later stage its content origin.						
	a)	Notarization	b)	Proof	c)	Nonrepudiation of Reception	d)
90.	_____ secret keys and possibly other data are to be kept confidential while transmitted or stored						
	a)	Data Confidentiality	b)	Modification Detection	c)	Replay Detection	d)
91.	_____ is to counter the active threat of unauthorized modification of data items						
	a)	Data Confidentiality	b)	Modification Detection	c)	Replay Detection	d)
92.	_____ is to counter unauthorized duplication of data items.						
	a)	Data Confidentiality	b)	Modification Detection	c)	Replay Detection	d)
93.	_____ is required that the response to a challenge message is prompt and does not allow for playback of some authentic response message by an impersonator.						
	a)	Data Confidentiality	b)	Modification Detection	c)	Replay Detection	d)
94.	_____ It involves storing, safeguarding, and activating keys						
	a)	key management	b)	key distribution	c)	Key Replacement	d)
95.	_____ refers to the procedures by which keys are securely provided to parties legitimately asking for them.						
	a)	key management	b)	key distribution	c)	Key Replacement	d)
96.	The _____ private key is used to decrypt cipher text in asymmetric cryptosystem						
	a)	Sender	b)	Receiver	c)	Both A & B	d)
97.	The _____ key is used to encrypt plain text in asymmetric cryptosystem						
	a)	public	b)	private	c)	Both A & B	d)
98.	_____ is the science of code breaking in a cryptographic schema						
	a)	Cryptography	b)	Cryptanalysis	c)	Cryptology	d)



99.	_____ is the study of encryption principles and methods.							
	a)	Cryptography	b)	Cryptanalysis	c)	Cryptology	d)	None
100.	_____ is the field of both cryptography and cryptanalysis.							
	a)	Cryptography	b)	Cryptanalysis	c)	Cryptology	d)	None
101.	In cryptography system, _____ is the original message before being transmitted							
	a)	Cipher text	b)	encrypted	c)	Plain text	d)	None
102.	In Caesar cipher, shifting a letter by +3 positions means that							
	a)	A becomes D, B becomes E, C becomes F, and so on						
	b)	A becomes X, B becomes Y, C becomes Z, and so on						
	c)	A becomes C, B becomes F, C becomes I, and so on						
	d)	A becomes B, B becomes C, C becomes D, and so on						
103.	Type of firewall can decide what services to perform on the user's behalf in accordance with its available knowledge							
	a)	Screening routers	b)	Stateful inspection	c)	Guards	d)	Application proxies
104.	It perform a simple pattern-matching to report situations that corresponding to a known attack type							
	a)	Normal based	b)	Signature-based IDS	c)	Heuristic IDS	d)	Honey pot
105.	It implies overhearing without expending any extra effort							
	a)	Eavesdrop	b)	Wiretap	c)	Spoofing	d)	Session hijacking
106.	Triple-DES with two keys id defined by $C= Ek_3(Dk_2(Ek_1(P)))$, if K_1 equal K_3 , the triple-DES can be considered							
	a)	Double DES	b)	Triple DES	c)	Single DES	d)	Useless encryption
107.	Which of the following isn't a design goal of firewalls							
	a)	Always invoked	b)	Tamper-proof	c)	Complex code	d)	Simple functionality
108.	It tracks the sequence of packets and conditions from one packet to another to thwart an attack							
	a)	Screening routers	b)	Stateful inspection	c)	Application proxies	d)	Guards
109.	A sender should not be able falsely to deny later that he/she sent a message							
	a)	Integrity	b)	Confidentiality	c)	Privacy	d)	Non-repudiation
110.	Physical OS separation approach _____ the resource sharing and _____ the security level							
	a)	Minimizes , Maximizes	b)	Maximizes , Maximizes	c)	Maximizes , Minimizes	d)	Minimizes , Minimizes
111.	Physical OS separation approach _____ the security level and _____ the resource sharing							
	a)	Minimizes , Maximizes	b)	Maximizes , Minimizes	c)	Minimizes, Minimizes	d)	Maximizes , Maximizes
112.	Kerberos system is considered _____ party _____ way authentication technique							
	a)	Two , One	b)	Two , Two	c)	Three , Two	d)	Three , One
113.	Digital signature provides the following characteristics expect _____							
	a)	Message authentication	b)	Receiver authentication	c)	Sender authentication	d)	Message encryption



114.	_____ Is used for managing audit trails and highlighting user violation of policy or normal activity							
	a)	Firewall	b)	IDS	c)	Antivirus	d)	Virus scanner
115.	Correction system configuration error is a function of _____							
	a)	Heuristic-based IDS	b)	Personal firewall	c)	Proxy	d)	Signature-based IDS
116.	Denning's IDS model builds heuristic decision model based on tracing _____							
	a)	Actions and it's initiator	b)	Timestamp of each action	c)	Exception condition of each action	d)	All
117.	Cyber security rules for daily practicing are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Administration controls	d)	Physical controls
118.	_____ means that some unauthorized party (person, program) has gained access to data							
	a)	Interception	b)	Modification	c)	Interruption	d)	Fabrication
119.	Legal and ethical controls are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Policy controls	d)	Physical controls
120.	A virus Is designed to hide itself from detection by antivirus software							
	a)	Brain virus	b)	Boot sector virus	c)	Polymorphic virus	d)	Trojan horse
121.	_____ record an intruder's actions and even attempt to trace who the attacker is from actions, packet data, or connection							
	a)	ISS scanner	b)	Nessus	c)	Honey pot	d)	tripwire
122.	Which of the following is/are considered a candidate IDS profile?							
	a)	Logon and session activity	b)	Command or program usage	c)	File access activity	d)	All
123.	_____ used to protect the networks hosts from an outside address							
	a)	Firewall	b)	IDS	c)	Antivirus	d)	Virus scanner
124.	Denning's IDS model is example of _____							
	a)	Host based IDS	b)	Signature based IDS	c)	Anomaly based IDS	d)	Hybrid IDS
125.	In which different processes use different physical objects, such as separate printers for output requiring different levels of security							
	a)	Temporal separation	b)	Logical separation	c)	Physical separation	d)	Cryptographic separation
126.	Steganography uses _____ in graphic image to hide important data							
	a)	LSB	b)	HSB	c)	MSB	d)	NSB
127.	_____ Controls access on the basis of packet address or specific transport protocol type							
	a)	Stateful inspection	b)	Screening routers	c)	Application proxies	d)	Guards
128.	A firewall system characterized by being two-headed device is							
	a)	Screening routers	b)	Stateful inspection	c)	Application proxies	d)	Guards
129.	_____ Builds a model of acceptable behavior and flag exceptions to that model							
	a)	Normal based	b)	Signature-based IDS	c)	Heuristic IDS	d)	Honey pot



130.	It attracts the intruder, record his actions, and even attempt to trace who the attacker is from action, packet data, or connections.							
	a)	Heuristic IDS	b)	Honey pot	c)	Normal Based	d)	Signature-based IDS
131.	Intercepting and carrying on a session begun by another entity is known as _____ attack							
	a)	Eavesdrop	b)	Wiretap	c)	Spoofing	d)	Session hijacking
132.	_____ this form of protection limits not just the access to an object but the use made of that object after it has been access							
	a)	Share via access limitation	b)	Share all or share nothing	c)	Share Via capabilities	d)	Limit use of an object
133.	_____ is when one host pretends to be another.							
	a)	Eavesdrop	b)	Spoofing	c)	Wiretap	d)	Session hijacking
134.	With this form of protection, the owner of an object declares it to be public or private							
	a)	Share via access limitation	b)	Share all or share nothing	c)	Don't protect model	d)	Isolate mode
135.	_____ in which processes conceal their data and computations in such a way that they are unintelligible to outside processes							
	a)	Temporal Separation	b)	Logical Separation	c)	Physical separation	d)	Cryptographic separation
136.	_____ Change control on the OS instructions, when a computer is started							
	a)	Memory resident virus	b)	Boot sector virus	c)	One time execution	d)	All
137.	When OS runs the applications of different security requirements under illusion that no other processes exist							
	a)	Logical separation	b)	Physical separation	c)	Temporal separation	d)	Space separation
138.	_____ perform simple pattern-matching based on logon and session activity to detect attack							
	a)	Heuristic -based IDS	b)	Personal Firewall	c)	Proxy	d)	Signature-based IDS
139.	Firewall is an executable code runs on a dedicated device to protect inside from outside networks							
	a)	Firewall	b)	IDS	c)	Antivirus	d)	Virus scanner
140.	A company wants to block the connection between its LAN and "youtube ", simply they can apply _____							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls
141.	A company wants to set up an online price list so that outsiders can see the products and prices offered. It wants to be sure that no outsider can change the prices or product list, the outsiders can access only the price list, not any of the more sensitive files stored inside.							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls
142.	A company wants to allow its employees to use e-mail up to a limit of so many messages or so many characters of e-mail in the last so many days, simply they can apply _____							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls



143.	Which of the following is best description of Stateful inspection firewall working principle?							
	a)	It can see address and services protocol type	b)	It can see full data portion of packet	c)	It can see either address or data	d)	It can see full text of communication
144.	What is true about IDS?							
	a)	it could be stand-alone device attached to the network to monitor traffic throughout that network						
	b)	Honey pot is considered an example for IDS						
	c)	Regarding anomaly-based IDS, admin can mark a flagged behavior as acceptable or not.						
	d)	All						
145.	_____ means intercepting and carrying on a session begun by another entity.							
	a)	Masquerade	b)	Man-in-the-Middle Attack	c)	Session hijacking	d)	Impersonation
146.	Which of the following can describe the function of Firewall system?							
	a)	To encrypt all network traffic to prevent eavesdropping						
	b)	To block all incoming network traffic except for authorized sources						
	c)	To prevent all unauthorized activity of networks' internal users						
	d)	To monitor internal network traffic and detect potential security breaches						
147.	Denning's model is an example for _____							
	a)	heuristic based IDS	b)	signature based IDS	c)	Virus Scan software	d)	Firewall model
148.	Which profile is recommended for auditing?							
	a)	logon and session activity	b)	command or program usage	c)	file access activity	d)	All
149.	When OS runs different process of different security requirements on the same time with shared resources under illusion that no other processes exist							
	a)	Cryptographic separation	b)	Temporal separation	c)	logical separation	d)	Physical separation
150.	if different processes can run concurrently unaware of the presence of each other, and each process has its own address space, files, and other objects. This OS level of sharing is categorized as:							
	a)	Isolation level	b)	Share via capabilities	c)	Share via access limitations	d)	Share all or share nothing
151.	What is TRUE about Kerberos authentication protocol?							
	a)	It is a two-party, two-way authentication protocol						
	b)	It is a two-party, three-way authentication protocol						
	c)	It is a three-party, two-way authentication protocol						
	d)	It is a three-party, three-way authentication protocol						
152.	The ticket issued to the user by Kerberos protocol includes the following data							
	a)	User login details	b)	Time-date stamp	c)	User privilege on the accessed application	d)	All are true



153.	What is a common example of a man-in-the-middle attack							
	a)	Trojan horse	b)	DDoS attack	c)	Wi-Fi eavesdropping	d)	SQL injection
154.	What is true about IDS?							
	a)	It could be stand-alone device attached to the network to monitor traffic throughout that network						
	b)	Honey pot is considered an example for IDS						
	c)	Regarding anomaly-based IDS, admin can mark a flagged behavior as acceptable or not						
	d)	All						
155.	Which of the following is best description of stateful inspection firewall working principle?							
	a)	It can see address and services protocol type						
	b)	it can see either address or data						
	c)	it can see full portion of packet						
	d)	It can see full text of communication						
156.	Which of the following best describes a Denial-of-service (DoS: attack)?							
	a)	An attack that locks users out of their accounts						
	b)	An attack that forces a system to shut down by overwhelming it with traffic						
	c)	An attack that steals sensitive information						
	d)	An attack that exploits vulnerabilities in cryptographic algorithms						
157.	The Feistel function F-function: in DES Includes which major operation?							
	a)	Substitution, Expansion, Permutation, XOR with round key						
	b)	Expansion, Substitution, Permutation, XOR with round key						
	c)	Expansion, Permutation, Substitution, XOR with round key						
	d)	XOR with round key, Expansion, Substitution, Permutation						
158.	What is a disadvantage of Rail Fence cipher compared to other transposition ciphers?							
	a)	it is more complex to implement						
	b)	it has a smaller key space						
	c)	it is slower in encrypting and decrypting						
	d)	It is more vulnerable to attacks						
159.	What distinguishes a trojan horse from a virus?							
	a)	Trojans cannot execute on their own						
	b)	Trojans spread through email attachments, while viruses spread via networks						
	c)	Trojans define themselves as legitimate software, while viruses attach to files						
	d)	Trojans self-replicate, whereas viruses do not						



160.	What is the purpose of S-boxes in DES?						
	a)	To compress data					
	b)	To provide confusion by replacing input bits with different bits					
	c)	To generate subkeys					
	d)	To provide permutation					
161.	Which of the following best describes how transposition ciphers provide security?						
	a)	By completely replacing the original letter with different characters					
	b)	By making frequency analysis difficult through letter rearrangement					
	c)	By requiring a secret key that cannot be guessed					
	d)	By using a complex mathematical function to generate ciphertext					
162.	Which of the following statements about transposition ciphers is true?						
	a)	They require a key that determines letter substitution					
	b)	They are always more secure than substitution ciphers					
	c)	They change the letters in the plaintext					
	d)	They keep the original plaintext letters but rearrange their order					
163.	What is the main purpose of encryption in computer security?						
	a)	To protect data from unauthorized access					
	b)	To speed up data transmission					
	c)	To create backup copies of data					
	d)	To hide system vulnerabilities					
164.	What is buffer overflow?						
	a)	A type of firewall					
	b)	A vulnerability where data overflows into adjacent memory locations					
	c)	A secure threat for data storage					
	d)	A type of ransomware attack					
165.	Which of the following is NOT an encryption algorithm?						
	a)	RSA	b)	AES	c)	DSS	d)
166.	How many subkeys are generated from the main key in DES?						
	a)	8	b)	16	c)	32	d)
167.	What type of attack involves intercepting and modifying communication between two parties without their knowledge?						
	a)	Phishing	b)	Denial-of-Service	c)	Brute-Force	d)



168.	What is a security threat?											
	a)	A program that prevents hacking attempts										
	b)	Any event or action that could cause damage to a system or data										
	c)	A measure used to warn user										
	d)	Any action that improves system security										
169.	_____ runs on a single workstation or client or host, to protect that one host											
	a)	Signature-based IDS		b)	Anomaly-based IDS		c)	Host-based IDS		d)	Network-based IDS	
170.	_____ is a stand-alone device attached to the network to monitor traffic throughout that network											
	a)	Signature-based IDS		b)	Anomaly-based IDS		c)	Host-based IDS		d)	Network-based IDS	
171.	Which type of IDS Monitors for deviations from normal network behavior and can detect previously unknown threats?											
	a)	Signature-based IDS		b)	Anomaly-based IDS (Heuristic IDS)		c)	Host-based IDS		d)	Network-based IDS	



True/False:

1.	Spoofing attack consist of replacing the valid source and/or destination IP address with false one	
2.	Bug is a mistake in interpreting a requirement or syntax error in a piece of code	
3.	For exchanging symmetric key among 10 users, we need 100 exchange operation	
4.	Erasure of program or data or malfunction of an OS file manager is considered fabrication	
5.	Validation error, domain error, and boundary condition violation are types of flaws	
6.	Erasure of program or data or malfunction of an OS file manager is considered Modification	
7.	Timely response is an expectation of availability security goal.	
8.	Verification of the user's identity is known as authorization process	
9.	Digital signatures provide the ability to verify both owner and content of message	
10.	The plain text must be broken into a block size of 128bit as an input of Triple DES	
11.	Polymorphic virus is a form of virus explicitly designed to hide itself from detection	
12.	A boot sector virus changes control on the OS instructions, when a computer is started	
13.	Playfair cipher rearranges the order of words in a message	
14.	Shift cipher applies a mathematical formula to each letter in a message that replaces each letter with a numerical value.	
15.	Flooding is an attack of availability	
16.	Authorization is mechanism used to confirm a user's identity	
17.	Fingerprint is used to verify the user as an example for something user has	
18.	Cryptographic separation is applied when OS runs different process of different security requirements on different time with separate resources	
19.	Trapdoor is a malicious software runs when a computer is started, it Changes control on the OS instructions	
20.	Program controls must include peer review steps	
21.	In context of Program controls, proof of program correctness is done based on check list	
22.	Code flaws include validation error or domain error	
23.	Heuristic IDS performs simple pattern-matching to report situation that corresponding to a known attack type	
24.	Eavesdrop Implies overhearing without expending any extra effort	
25.	The firewall can't be able to detect the signature of the attack if it was broken into multiple packets each have very short length	
26.	A personal firewall is an application program that runs on a user's workstation usually under his policy	
27.	Auditing is difficult when applying Packet filtering firewalls	
28.	A network-based IDS is installed on the network while a host-based IDS is installed on individual hosts.	



الاحتياط واجب

Choose the correct answer:

1.	Ensuring that computer assets are accessed only by authorized parties and authorized parties actually access the required assets is a goal of _____.							
	a)	Confidentiality	b)	Integrity	c)	Availability	d)	All of the above
2.	_____ is Ensuring that the data is accurate, consistent, and not tampered with.							
	a)	Confidentiality	b)	Integrity	c)	Availability	d)	All of the above
3.	Using systems like Git to track and prevent unapproved modifications in codebases achieves _____ of the data							
	a)	Confidentiality	b)	Integrity	c)	Availability	d)	All of the above
4.	_____ Is Ensuring that the systems and the data are accessible when needed.							
	a)	Confidentiality	b)	Integrity	c)	Availability	d)	All of the above
5.	Service is denied to unauthorized users is an example of data availability							
	a)	TRUE	b)	FALSE	c)		d)	
6.	When an asset of the system is destroyed or becomes unusable, this is called _____ and this is an attack on the _____.							
	a)	Interruption, availability	b)	Interception, availability	c)	modification, availability	d)	modification, integrity
7.	An unauthorized party only gains access to an asset this is called _____ and this is an attack on the _____.							
	a)	Interruption, availability	b)	Interception, confidentiality	c)	modification, availability	d)	modification, integrity
8.	An unauthorized party not only gains access to an asset but tampers with an asset this is called _____ and this is an attack on the _____.							
	a)	Interruption, availability	b)	Interception, confidentiality	c)	modification, availability	d)	modification, integrity
9.	“Who are you”? is an _____ question while “What are you allowed to do?” is an _____ question							
	a)	Authentication, Authorization	b)	Authorization, Authentication	c)	Authentication, Authentication	d)	Authorization, Authorization

